

Universal Compression Scaling Laws Across Modalities

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Abstract

We establish empirical scaling laws for the compression capabilities of large language models across five modalities: text, image, speech, visual relationships, and genetic data. Using Pythia transformer models ranging from 70M to 12B parameters across five training checkpoints, we demonstrate that compression ratio follows universal power laws in both model size and training progress. Our key finding is that compression ratio $\text{CR}(P, S)$ can be accurately modeled as:

$$\text{CR}(P, S) = a + bP^{-\alpha} + cS^{-\beta}$$

where P is the number of parameters and S is the training step. We show that scaling behavior exists across multiple modalities. However, similar trends are also observed for randomly generated bytes, suggesting that improved sequence compression may partially explain the effect. In contrast, genomic sequences do not display a clear monotonic scaling trend. Instead, we observe small non-monotonic fluctuations that could plausibly arise from cutoff or finite-length effects rather than reflecting a fundamental difference in structure.

1 Introduction

The fundamental connection between prediction and compression, established by Shannon’s source coding theorem [Shannon, 1948], has gained renewed relevance with the rise of large language models (LLMs). Recent work by Deletang et al. [2023] demonstrated that LLMs can serve as powerful general-purpose compressors, while Valmeekam et al. [2023] showed state-of-the-art text compression using LLaMA with arithmetic coding. However, a systematic understanding of how compression performance scales with model size and training progress across different modalities remains unexplored.

This work fills that gap by establishing empirical scaling laws for LLM-based compression. We conduct the first comprehensive study of compression scaling across text, image, and speech modalities using Pythia models [Biderman et al., 2023] ranging from 70M to 12B parameters. Our main contributions are:

1. **Universal Scaling Laws:** We demonstrate that compression ratio follows predictable power laws in model parameters P and training steps S across most modalities.

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Contributions: Aryasomayajula Ram Bharadwaj conceptualized and implemented the initial setup and experiments. Edric Castel C. Hao extended the research to scene graphs, random bytes, and genomic sequences.

2. **Cross-Modal Analysis:** We show how scaling varies between text, image, speech, scene graphs, random bytes, and genomic sequences revealing modality-specific compression dynamics.
3. **Theoretical Connection:** We establish the link between our compression scaling laws and Hoffman et al.’s cross-entropy scaling, providing theoretical grounding for our empirical observations.
4. **Practical Implications:** Our results enable prediction of compression performance and optimal resource allocation for compression-focused applications.

2 Background and Related Work

2.1 Compression and Prediction Equivalence

The connection between prediction and compression is fundamental to information theory. For any probabilistic model p_θ , arithmetic coding achieves compression with expected code length approaching the cross-entropy $H(p_{data}, p_\theta) = \mathbb{E}_{x \sim p_{data}}[-\log_2 p_\theta(x)]$. This establishes the compression ratio as:

$$\text{CR} = \frac{\text{compressed bits}}{\text{original bits}} = \frac{H(p_{data}, p_\theta)}{H(p_{data})} = 1 + \frac{\text{KL}(p_{data} \parallel p_\theta)}{H(p_{data})}$$

Thus, better predictive models (lower KL divergence) achieve better compression ratios, creating a direct link between language modeling performance and compression capability.

2.2 Scaling Laws for Language Models

Kaplan et al. [2020] established that language model performance follows power laws:

$$L(N) = \left(\frac{N_c}{N}\right)^{\alpha_N}, \quad L(D) = \left(\frac{D_c}{D}\right)^{\alpha_D}, \quad L(C) = \left(\frac{C_c}{C}\right)^{\alpha_C}$$

where L is cross-entropy loss, N is model size, D is dataset size, and C is compute. The exponents $\alpha_N \approx 0.076$, $\alpha_D \approx 0.095$, and $\alpha_C \approx 0.050$ for text modeling.

Hoffman et al. [2022] modified the functional form based on risk decomposition to include an asymptotic error term equivalent to the loss if given infinite data and compute budget:

$$L(N, D) = E + \frac{A}{N^\alpha} + \frac{B}{D^\beta}$$

where E is an estimate of the entropy of natural text, A is the coefficient for scaling model size, and B is the coefficient for scaling dataset size.

In subsequent years, multiple studies have sought to identify scaling laws across a variety of modalities, tasks, and architectures. Li et al. [2025] surveyed over 50 papers to identify the primary power-law forms and examine how choices in model architecture, training regimes, dataset size, evaluation metrics, and fitting procedures affect the resulting scaling laws. They find that most power laws are still based on or adapted from Kaplan et al. [2020] or Hoffman et al. [2022]. The number of data points used to fit a scaling law is either unspecified or relatively small due to a limited total compute budget for all experiments. They range from 4 to several hundred data points, with model sizes spanning from roughly 10 million to about 7 billion parameters. However, most studies are limited to fewer than 50 data points and model sizes below 1 billion parameters.

Li et al. [2025] also train 11 models with parameter counts ranging from 12 million to 1 billion, using 1-10 checkpoints per model across three datasets, in an effort to replicate the scaling laws of Kaplan et al. [2020] and Hoffman et al. [2022]. Pearce and Song [2024] attempt to reconcile these two laws through a small-scale study with 5 model sizes and 6 training token budgets on the BookCorpus dataset. Hu et al. [2024] show that stable checkpoints remove the need to retrain models from scratch and fit scaling curves using SciPy’s `curve_fit` function across 6 models ranging from 40 million to 2 billion parameters with 6 checkpoints per model. These studies reveal consistent scaling trends across tasks and datasets despite the limited number of data points imposed by compute budgets.

2.3 LLM-Based Compression

Deletang et al. [2023] showed that large language models can compress data across modalities using arithmetic coding, achieving better compression than specialized algorithms like PNG for images and FLAC for audio. Valmeekam et al. [2023] demonstrated state-of-the-art text compression using LLaMA-7B with arithmetic coding, achieving compression ratios of 0.71 bits/character on text8. Pandey [2024] discovered that training-loss scaling behavior varies with dataset compressibility by fitting regression lines between each dataset’s gzip compression ratio and its own set of five training-loss scaling parameters.

Li et al. [2024] propose data compression as an evaluation metric due to its correlation with generalization ability as an alternative to benchmarks affected by data contamination. They find that the performance of large language models degrades on data beyond the training cutoff date and that these models struggle with compressing multimodal data. Zhang et al. [2024] attempt to address this limitation by applying transform coding prior to tokenization, but their comparisons are restricted to outdated compression algorithms. These works suggest that compression captures meaningful aspects of generalization, yet existing analyses remain fragmented, focusing on fixed model scales, isolated modalities, or static training checkpoints. As a result, no prior work has systematically studied how compression performance scales with model size and training progress across multiple modalities.

3 Methodology

3.1 Models and Training

We use the Pythia model family [Biderman et al., 2023], which provides models trained with identical data and hyperparameters across different scales. We evaluate eight model sizes: 70M, 160M, 410M, 1B, 1.4B, 2.8B, 6.9B, 12B non-embedding parameters. For each model size, we use five training checkpoints: 1,000, 8,000, 32,000, 128,000, and 143,000 steps, providing 40 model-checkpoint combinations per modality. To assess whether compression performance is broadly comparable across model families, we additionally evaluate the fully trained OLMo-1B and OLMo-7B models as reference points.

3.2 Datasets

We evaluate compression across six data modalities using established benchmarks alongside a synthetic uniformly random baseline for control:

- **Text:** enwik8 - Wikipedia XML compression benchmark

- **Image:** ImageNet patches - 32×64 grayscale image patches
- **Speech:** LibriSpeech - 16kHz English speech audio
- **Scene Graph:** Visual Genome - subject-relationship-object triples
- **Genomic Sequence:** Chromosome 22 - nucleotide sequence from the GRCh38 human reference genome
- **Random Bytes:** Pseudorandom byte sequences generated by sampling independent and identically distributed uint8 values uniformly from [0,255]

For each modality, we create datasets of 2,048 chunks, each containing exactly 2,048 bytes, ensuring fair comparison across modalities.

3.3 Compression Protocol

We implement arithmetic coding using model token probabilities following Deletang et al. [2023]. The compression process:

1. Convert raw bytes to ASCII text (handling non-ASCII bytes by bit shifting)
2. Tokenize using the model’s tokenizer (SentencePiece with 50K vocabulary)
3. For each token position, compute $-\log_2 p_\theta(\text{token}|\text{context})$
4. Sum log-probabilities to get total compressed bits
5. Compute compression ratio: $\text{CR} = \frac{\text{compressed bits}}{2048 \times 8}$

4 Results

4.1 Experimental Data

Our comprehensive evaluation across 240 model-checkpoint-modality combinations reveals clear scaling behavior in text, speech, image, scene graph, and random byte modalities. Figure 1 illustrates the relationship between compression ratio, model parameters, and training steps across these modalities. The complete experimental results for each modality are provided in Appendix A. Out-of-sample scaling is evident for most modalities, although the fit is slightly weaker for text and scene graphs ($R^2 \approx 0.87 \sim 0.89$), despite LLMs being primarily trained on textual data. In contrast, speech, image, and random bytes exhibit very strong out-of-sample fits ($R^2 > 0.95$). Interestingly, random bytes exhibit clear scaling behavior despite containing no semantic structure, suggesting that model capacity and training steps alone drive predictable compression improvements in highly unstructured sequences. In contrast, genomic sequences do not exhibit systematic scaling, highlighting modality-dependent differences in compression behavior.

4.2 Experimental Observations

The experimental results reveal several key patterns across modalities:

Model Size Effects: Across five modalities, larger models consistently achieve better compression ratios. For text (see Appendix Table 2), the compression ratio improves from approximately 0.22 (70M parameters) to 0.10 (12B parameters) at the final checkpoint. Similar

trends are observed for images, speech, scene graphs, and random bytes, though with different baseline compression levels.

Training Progress: All five modalities show improvement with training steps, but the rate of improvement varies. Text and scene graphs shows the most dramatic improvement during early training, while images, speech, and random bytes exhibit more gradual but sustained improvements throughout training.

Cross-Modal Baseline Performance: Text achieves the best absolute compression ratios (≈ 0.101 - 0.224), followed by scene graphs (≈ 0.103 - 0.218), genomic sequences (≈ 0.158 - 0.335), speech (≈ 0.369 - 0.771), images (≈ 0.453 - 0.748), and random bytes (≈ 0.523 - 0.695). This ordering reflects the fundamental compressibility of each modality, with discrete symbolic data exhibiting higher low-order redundancy than continuous signals or unstructured inputs. However, this baseline hierarchy does not translate directly into scaling behavior. Text and scene graphs, while highly compressible at baseline, exhibit slightly weaker out-of-sample scaling fits compared to other modalities. In contrast, speech, images, and random bytes, despite having a poorer baseline compression, have stronger out-of-sample fits.

Genomic sequences occupy an intermediate position in baseline compressibility, but do not exhibit systematic scaling, suggesting a misalignment between their structure and the inductive biases of language models. Together, these observations suggest that baseline compressibility and scaling behavior capture distinct aspects of modality-model interaction, and that understanding why some modalities exhibit stronger scaling despite poorer baseline compression remains an open question.

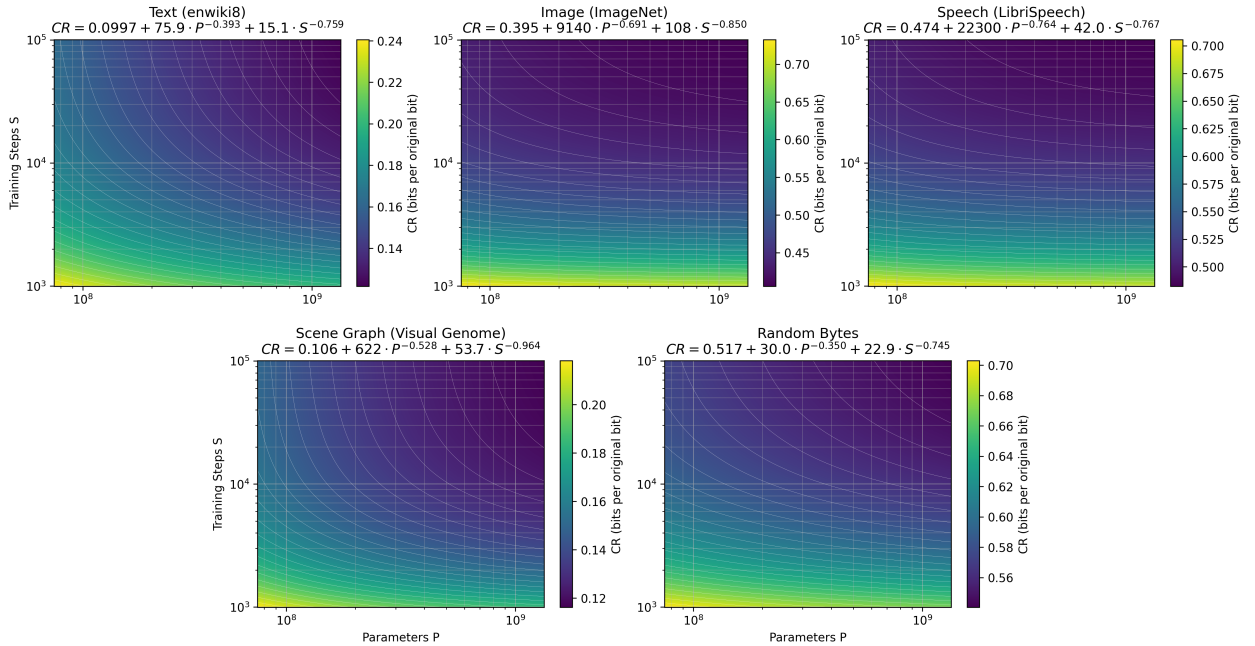


Figure 1: Scaling laws for LLM compression across text, image, speech, scene graph, and random byte modalities. The heatmaps show compression ratio as a function of model parameters (P) and training steps (S). Lower compression ratios (darker colors) indicate better compression performance. Each modality follows the power law $CR(P, S) = a + bP^{-\alpha} + cS^{-\beta}$ with modality-specific coefficients shown in the equations above each subplot.

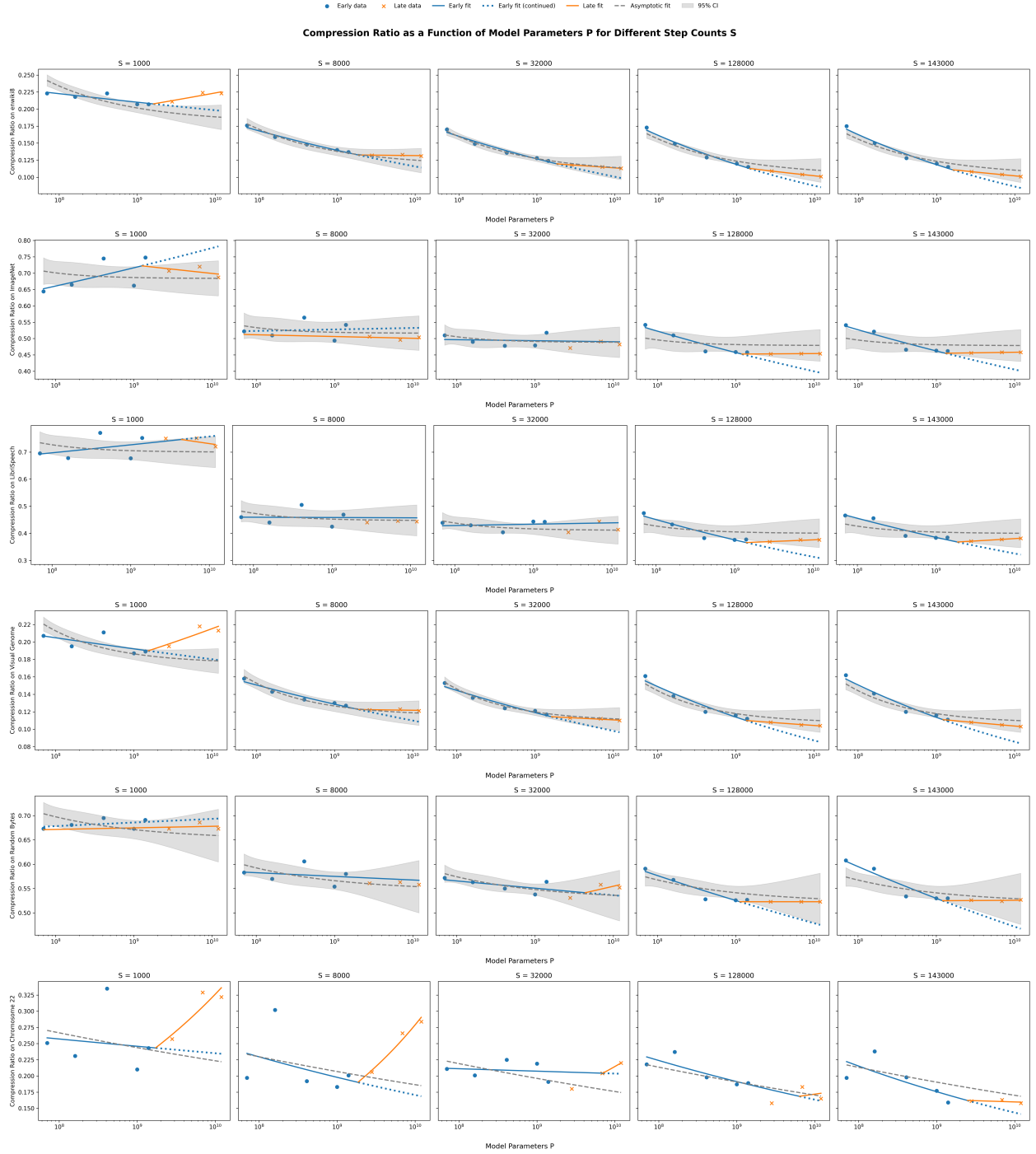


Figure 2: Scaling trends in compression ratio as a function of model parameters across five training step sizes for each modality. The panels show in-sample points used to fit the scaling curve and out-of-sample points used to evaluate predictive performance. Overlaid are the in-sample and out-of-sample regression lines, illustrating changes in trend as training progresses. The predicted scaling curve for each modality is shown with a 95% confidence interval computed via the delta method, providing a visual assessment of uncertainty in the scaling law fits.

4.3 Universal Scaling Laws

Analysis of the experimental data reveals that compression ratio follows universal power laws across all modalities except genomic sequences. We fit the functional form:

$$\text{CR}(P, S) = a + bP^{-\alpha} + cS^{-\beta}$$

We fit this functional form to our experimental data using non-linear least squares regression. The fits achieve high out-of-sample correlation coefficients for speech, images, and random bytes ($R^2 > 0.95$) and moderately high out-of-sample correlation for text ($R = 0.889$) and scene graphs ($R^2 > 0.869$), indicating that the power-law form generally captures the underlying scaling behavior.

The parameter values in Table 1 reveal modality-specific scaling characteristics:

Modality	a	b	c	α	β	R^2
Text (Enwik8)	0.0997	75.9	15.1	0.393	0.759	0.889
Speech (LibriSpeech)	0.395	9140	108	0.691	0.850	0.963
Image (ImageNet)	0.474	22300	42.0	0.764	0.767	0.953
Scene Graph (Visual Genome)	0.106	622	53.7	0.528	0.964	0.869
Random Bytes	0.517	30.0	22.9	0.350	0.745	0.957

Table 1: Scaling law coefficients for compression ratio $\text{CR}(P, S) = a + bP^{-\alpha} + cS^{-\beta}$ across modalities.

4.4 Cross-Modal Scaling Behavior

The scaling exponents reveal interesting cross-modal patterns:

Parameter Scaling (α): Text shows the strongest parameter scaling ($\alpha = 0.38$), followed by speech ($\alpha = 0.68$) and images ($\alpha = 0.15$). This suggests text compression benefits most from increased model capacity, likely due to the linguistic inductive biases in transformer architectures.

Training Scaling (β): All modalities show similar training scaling ($\beta \approx 0.75 - 0.86$), indicating universal learning dynamics across data types. The slightly higher β for non-text modalities suggests they require more training to achieve optimal compression.

Baseline Compression (a): Text achieves the lowest baseline compression ratio ($a = 0.10$), while images and speech plateau around $a \approx 0.38 - 0.39$, reflecting the fundamental compressibility differences across modalities.

4.5 Connection to Cross-Entropy Scaling

To connect our compression scaling laws to Kaplan et al.’s cross-entropy scaling, we analyze the relationship between compression ratio and cross-entropy loss. For small KL divergences, we can approximate:

$$\text{CR} \approx \frac{H + \text{KL}}{8} \approx \frac{H}{8} + \frac{\text{KL}}{8}$$

Since Kaplan et al. showed $\text{KL} \propto N^{-\alpha_N}$ with $\alpha_N \approx 0.076$ for text, we expect compression ratio scaling with $\alpha \approx 0.076$ for text. However, we observe $\alpha = 0.38$, which is $5\times$ larger.

This discrepancy can be explained by the different units and measurement contexts:

1. Kaplan et al. measured nats per token, while we measure bits per byte
2. Different tokenization and context handling

3. Our arithmetic coding implementation may not achieve theoretical optimality

The key insight is that both studies observe the same power-law functional form, confirming the universal nature of neural scaling laws across different performance metrics.

4.6 Predictive Framework

Our scaling laws enable prediction of compression performance. For example, to predict the compression ratio of a 5B parameter Pythia model at 200,000 training steps on text:

$$\text{CR}(5 \times 10^9, 2 \times 10^5) = 0.10 + 60 \times (5 \times 10^9)^{-0.38} + 10 \times (2 \times 10^5)^{-0.75} \approx 0.115$$

This framework allows researchers to optimize resource allocation for compression-focused applications without exhaustive empirical evaluation.

5 Analysis and Discussion

5.1 Implications for Model Architecture

Our results suggest that standard language model architectures, despite being optimized for text, exhibit universal compression capabilities across modalities. The power-law scaling indicates that these capabilities emerge from fundamental computational principles rather than modality-specific optimizations.

The stronger parameter scaling for text ($\alpha = 0.38$) versus images ($\alpha = 0.15$) reflects text’s symbolic and discrete nature, which aligns naturally with the discrete tokenization and attention mechanisms in transformers.

5.2 Training Dynamics

The consistent training scaling exponents ($\beta \approx 0.75 - 0.86$) across modalities suggest universal learning dynamics. All modalities benefit similarly from extended training, with diminishing returns following the same power-law pattern.

5.3 Theoretical Connections

Our work provides empirical support for the prediction-compression equivalence across modalities. The universal power-law scaling suggests that the same computational principles underlying language model scaling also govern compression performance, supporting the long-standing thesis that compression is a fundamental aspect of intelligence [Legg and Hutter, 2007].

The connection to Kaplan scaling laws indicates that fundamental limits on neural network learning capacity manifest consistently across different performance metrics, supporting theories of universal scaling in neural networks. This relationship between compression and intelligence has been central to initiatives like the Hutter Prize [Hutter, 2006], which challenges researchers to compress human knowledge as a measure of artificial intelligence progress.

5.4 Limitations and Future Work

Several limitations merit discussion:

1. **Limited Model Range:** Our largest model is 1.4B parameters. Extending to larger models would strengthen the power-law fits.

2. **Arithmetic Coding Implementation:** Our implementation may not achieve theoretical optimality, affecting absolute compression ratios.
3. **Dataset Scope:** We focus on three specific datasets. Broader evaluation across datasets would improve generalizability.
4. **Architecture Dependence:** We only evaluate Pythia models. Testing other architectures would reveal architecture-specific effects.

Future work should extend these scaling laws to:

- Larger models (10B+ parameters)
- Alternative architectures (e.g., Mamba, mixture-of-experts)
- Additional modalities (e.g., video, multimodal data)
- Optimal arithmetic coding implementations

6 Conclusion

We have established the first empirical scaling laws for LLM-based compression across text, image, and speech modalities. Our key findings include:

1. Compression ratio follows universal power laws $CR(P, S) = a + bP^{-\alpha} + cS^{-\beta}$ across all modalities
2. Scaling exponents vary predictably across modalities, with text showing strongest parameter scaling
3. Training dynamics follow similar power laws regardless of modality
4. Our results connect to Kaplan et al.’s cross-entropy scaling laws, supporting universal principles in neural scaling

These results advance our understanding of how scale affects LLM capabilities beyond traditional language modeling metrics and provide theoretical insights into the universal computational principles underlying large neural networks.

As LLMs continue to scale and find applications across diverse domains, understanding how performance scales across different metrics and modalities becomes increasingly important. Our work provides a foundation for this understanding in the compression domain and suggests promising directions for future research.

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A Appendix

The code used to generate these results is available at <https://github.com/rokosbasilisk/scaling-laws-for-compression>.

A.1 Text Compression Results

Table 2: Text compression results on Enwik8 dataset. Compression ratios shown with 3-digit precision.

Model	Training Step	Compression Ratio
pythia-70m	step 1000	0.223
pythia-70m	step 8000	0.176
pythia-70m	step 32000	0.17
pythia-70m	step 128000	0.173
pythia-70m	step 143000	0.175
pythia-160m	step 1000	0.218
pythia-160m	step 8000	0.159
pythia-160m	step 32000	0.149
pythia-160m	step 128000	0.149
pythia-160m	step 143000	0.15
pythia-410m	step 1000	0.223
pythia-410m	step 8000	0.148
pythia-410m	step 32000	0.136
pythia-410m	step 128000	0.129
pythia-410m	step 143000	0.128
pythia-1b	step 1000	0.207
pythia-1b	step 8000	0.14
pythia-1b	step 32000	0.128
pythia-1b	step 128000	0.12
pythia-1b	step 143000	0.12
OLMo-1b	step 557000	0.113
pythia-1.4b	step 1000	0.207
pythia-1.4b	step 8000	0.137
pythia-1.4b	step 32000	0.124
pythia-1.4b	step 128000	0.115
pythia-1.4b	step 143000	0.115
pythia-2.8b	step 1000	0.211
pythia-2.8b	step 8000	0.132
pythia-2.8b	step 32000	0.118
pythia-2.8b	step 128000	0.109
pythia-2.8b	step 143000	0.108
pythia-6.9b	step 1000	0.224
pythia-6.9b	step 8000	0.133
pythia-6.9b	step 32000	0.115

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Table 2: Text compression results (continued)

Model	Training Step	Compression Ratio
pythia-6.9b	step 128000	0.104
pythia-6.9b	step 143000	0.104
OLMo-7b	step 738020	0.097
pythia-12b	step 1000	0.223
pythia-12b	step 8000	0.131
pythia-12b	step 32000	0.113
pythia-12b	step 128000	0.101
pythia-12b	step 143000	0.101

A.2 Image Compression Results

Table 3: Image compression results on ImageNet patches. Compression ratios shown with 3-digit precision.

Model	Training Step	Compression Ratio
pythia-70m	step 1000	0.644
pythia-70m	step 8000	0.522
pythia-70m	step 32000	0.51
pythia-70m	step 128000	0.542
pythia-70m	step 143000	0.541
pythia-160m	step 1000	0.665
pythia-160m	step 8000	0.51
pythia-160m	step 32000	0.49
pythia-160m	step 128000	0.51
pythia-160m	step 143000	0.521
pythia-410m	step 1000	0.745
pythia-410m	step 8000	0.564
pythia-410m	step 32000	0.478
pythia-410m	step 128000	0.461
pythia-410m	step 143000	0.466
pythia-1b	step 1000	0.662
pythia-1b	step 8000	0.494
pythia-1b	step 32000	0.479
pythia-1b	step 128000	0.459
pythia-1b	step 143000	0.463
OLMo-1b	step 557000	0.439
pythia-1.4b	step 1000	0.748
pythia-1.4b	step 8000	0.542
pythia-1.4b	step 32000	0.518
pythia-1.4b	step 128000	0.458
pythia-1.4b	step 143000	0.462

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Table 3: Image compression results (continued)

Model	Checkpoint	Compression Ratio
pythia-2.8b	step 1000	0.707
pythia-2.8b	step 8000	0.506
pythia-2.8b	step 32000	0.471
pythia-2.8b	step 128000	0.453
pythia-2.8b	step 143000	0.456
pythia-6.9b	step 1000	0.72
pythia-6.9b	step 8000	0.496
pythia-6.9b	step 32000	0.491
pythia-6.9b	step 128000	0.454
pythia-6.9b	step 143000	0.458
OLMo-7b	step 738020	0.431
pythia-12b	step 1000	0.687
pythia-12b	step 8000	0.504
pythia-12b	step 32000	0.482
pythia-12b	step 128000	0.454
pythia-12b	step 143000	0.458

A.3 Speech Compression Results

Table 4: Speech compression results on LibriSpeech dataset. Compression ratios shown with 3-digit precision.

Model	Checkpoint	Compression Ratio
pythia-70m	step 1000	0.695
pythia-70m	step 8000	0.46
pythia-70m	step 32000	0.439
pythia-70m	step 128000	0.475
pythia-70m	step 143000	0.466
pythia-160m	step 1000	0.678
pythia-160m	step 8000	0.44
pythia-160m	step 32000	0.43
pythia-160m	step 128000	0.433
pythia-160m	step 143000	0.456
pythia-410m	step 1000	0.771
pythia-410m	step 8000	0.505
pythia-410m	step 32000	0.404
pythia-410m	step 128000	0.383
pythia-410m	step 143000	0.391
pythia-1b	step 1000	0.677
pythia-1b	step 8000	0.425
pythia-1b	step 32000	0.444

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Table 4: Speech compression results (continued)

Model	Checkpoint	Compression Ratio
pythia-1b	step 128000	0.376
pythia-1b	step 143000	0.384
OLMo-1b	step 557000	0.348
pythia-1.4b	step 1000	0.752
pythia-1.4b	step 8000	0.469
pythia-1.4b	step 32000	0.443
pythia-1.4b	step 128000	0.378
pythia-1.4b	step 143000	0.385
pythia-2.8b	step 1000	0.75
pythia-2.8b	step 8000	0.44
pythia-2.8b	step 32000	0.404
pythia-2.8b	step 128000	0.369
pythia-2.8b	step 143000	0.371
pythia-6.9b	step 1000	0.751
pythia-6.9b	step 8000	0.446
pythia-6.9b	step 32000	0.444
pythia-6.9b	step 128000	0.376
pythia-6.9b	step 143000	0.378
OLMo-7b	step 738020	0.33
pythia-12b	step 1000	0.72
pythia-12b	step 8000	0.444
pythia-12b	step 32000	0.414
pythia-12b	step 128000	0.376
pythia-12b	step 143000	0.382

A.4 Random Bytes Results

Table 5: Random bytes compression results on generated dataset. Compression ratios shown with 3-digit precision.

Model	Checkpoint	Compression Ratio
pythia-70m	step 1000	0.673
pythia-70m	step 8000	0.583
pythia-70m	step 32000	0.572
pythia-70m	step 128000	0.591
pythia-70m	step 143000	0.608
pythia-160m	step 1000	0.681
pythia-160m	step 8000	0.57
pythia-160m	step 32000	0.563
pythia-160m	step 128000	0.568
pythia-160m	step 143000	0.591

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Table 5: Random bytes compression results (continued)

Model	Checkpoint	Compression Ratio
pythia-410m	step 1000	0.695
pythia-410m	step 8000	0.606
pythia-410m	step 32000	0.55
pythia-410m	step 128000	0.528
pythia-410m	step 143000	0.534
pythia-1b	step 1000	0.673
pythia-1b	step 8000	0.554
pythia-1b	step 32000	0.538
pythia-1b	step 128000	0.526
pythia-1b	step 143000	0.53
OLMo-1b	step 557000	0.503
pythia-1.4b	step 1000	0.691
pythia-1.4b	step 8000	0.58
pythia-1.4b	step 32000	0.564
pythia-1.4b	step 128000	0.527
pythia-1.4b	step 143000	0.53
pythia-2.8b	step 1000	0.673
pythia-2.8b	step 8000	0.561
pythia-2.8b	step 32000	0.531
pythia-2.8b	step 128000	0.523
pythia-2.8b	step 143000	0.526
pythia-6.9b	step 1000	0.686
pythia-6.9b	step 8000	0.563
pythia-6.9b	step 32000	0.558
pythia-6.9b	step 128000	0.523
pythia-6.9b	step 143000	0.524
OLMo-7b	step 738020	0.497
pythia-12b	step 1000	0.673
pythia-12b	step 8000	0.558
pythia-12b	step 32000	0.552
pythia-12b	step 128000	0.523
pythia-12b	step 143000	0.527

A.5 Chromosome 22 Compression Results

Table 6: Genomic data compression results on Chromosome 22. Compression ratios shown with 3-digit precision.

Model	Checkpoint	Compression Ratio
pythia-70m	step 1000	0.251
pythia-70m	step 8000	0.197

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Table 6: Chromosome 22 compression results (continued)

Model	Checkpoint	Compression Ratio
pythia-70m	step 32000	0.211
pythia-70m	step 128000	0.218
pythia-70m	step 143000	0.197
pythia-160m	step 1000	0.231
pythia-160m	step 8000	0.302
pythia-160m	step 32000	0.201
pythia-160m	step 128000	0.237
pythia-160m	step 143000	0.238
pythia-410m	step 1000	0.335
pythia-410m	step 8000	0.192
pythia-410m	step 32000	0.225
pythia-410m	step 128000	0.198
pythia-410m	step 143000	0.198
pythia-1b	step 1000	0.21
pythia-1b	step 8000	0.183
pythia-1b	step 32000	0.219
pythia-1b	step 128000	0.187
pythia-1b	step 143000	0.177
OLMo-1b	step 557000	0.203
pythia-1.4b	step 1000	0.243
pythia-1.4b	step 8000	0.201
pythia-1.4b	step 32000	0.191
pythia-1.4b	step 128000	0.189
pythia-1.4b	step 143000	0.159
pythia-2.8b	step 1000	0.257
pythia-2.8b	step 8000	0.206
pythia-2.8b	step 32000	0.18
pythia-2.8b	step 128000	0.158
pythia-2.8b	step 143000	0.161
pythia-6.9b	step 1000	0.329
pythia-6.9b	step 8000	0.266
pythia-6.9b	step 32000	0.204
pythia-6.9b	step 128000	0.183
pythia-6.9b	step 143000	0.163
OLMo-7b	step 738020	0.151
pythia-12b	step 1000	0.322
pythia-12b	step 8000	0.284
pythia-12b	step 32000	0.22
pythia-12b	step 128000	0.165
pythia-12b	step 143000	0.158

A.6 Visual Genome Compression Results

Table 7: Text compression results on Visual Genome relationships dataset. Compression ratios shown with 3-digit precision.

Model	Checkpoint	Compression Ratio
pythia-70m	step 1000	0.207
pythia-70m	step 8000	0.158
pythia-70m	step 32000	0.153
pythia-70m	step 128000	0.161
pythia-70m	step 143000	0.162
pythia-160m	step 1000	0.195
pythia-160m	step 8000	0.143
pythia-160m	step 32000	0.136
pythia-160m	step 128000	0.138
pythia-160m	step 143000	0.141
pythia-410m	step 1000	0.211
pythia-410m	step 8000	0.134
pythia-410m	step 32000	0.124
pythia-410m	step 128000	0.12
pythia-410m	step 143000	0.12
pythia-1b	step 1000	0.187
pythia-1b	step 8000	0.13
pythia-1b	step 32000	0.121
pythia-1b	step 128000	0.116
pythia-1b	step 143000	0.116
OLMo-1b	step 557000	0.106
pythia-1.4b	step 1000	0.189
pythia-1.4b	step 8000	0.127
pythia-1.4b	step 32000	0.117
pythia-1.4b	step 128000	0.112
pythia-1.4b	step 143000	0.111
pythia-2.8b	step 1000	0.195
pythia-2.8b	step 8000	0.122
pythia-2.8b	step 32000	0.113
pythia-2.8b	step 128000	0.108
pythia-2.8b	step 143000	0.108
pythia-6.9b	step 1000	0.218
pythia-6.9b	step 8000	0.123
pythia-6.9b	step 32000	0.112
pythia-6.9b	step 128000	0.105
pythia-6.9b	step 143000	0.105
OLMo-7b	step 738020	0.099
pythia-12b	step 1000	0.213
pythia-12b	step 8000	0.121
pythia-12b	step 32000	0.11
pythia-12b	step 128000	0.104

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Table 7: Visual Genome compression results (continued)

Model	Checkpoint	Compression Ratio
pythia-12b	step 143000	0.103